

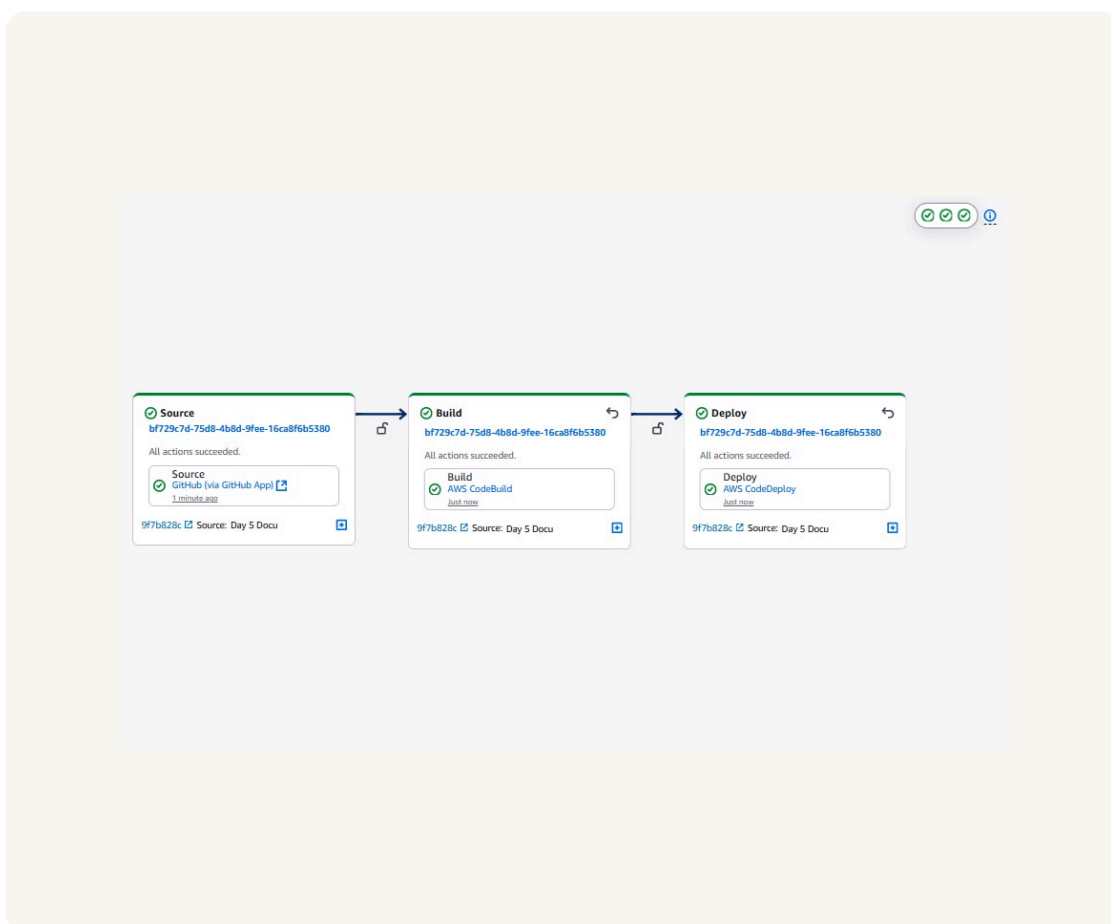


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Build a CI/CD Pipeline with AWS



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Introducing Today's Project!

In this project, I will demonstrate how to setup a CI/CD Pipeline in AWS. I'm doing this project to learn how to automate my deployment process in AWS.

Key tools and concepts

Services I used were CodeBuild, CodeDeploy, CodePipeline. Key concepts I learnt include the relationship between Github(Source), CodeBuild(Builder) and CodeDeploy(Deployer).

Project reflection

This project took me approximately 1 hour. The most challenging part was redoing the earlier steps before reaching CodePipeline step. It was most rewarding to see the CI/CD Pipeline work and seeing my webapp updating in real time.

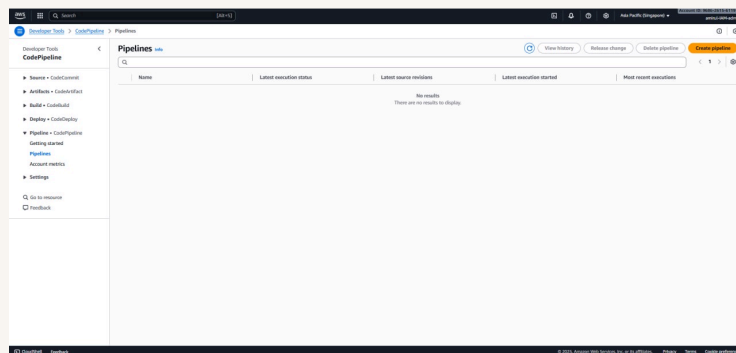


Starting a CI/CD Pipeline

AWS CodePipeline is a service that automatically triggers a build in CodeBuild and deploy using CodeDeploy whenever there is a new push to my Github Repo.

CodePipeline offers different execution modes based on how devs want to deal with multiple runs of the same pipeline. I chose Superseded as it cancels the other runs and only focus on the latest run. Other options include Queued and Parallel.

A service role gets created automatically during setup so that it gets permission to access other AWS resources while running the pipeline.

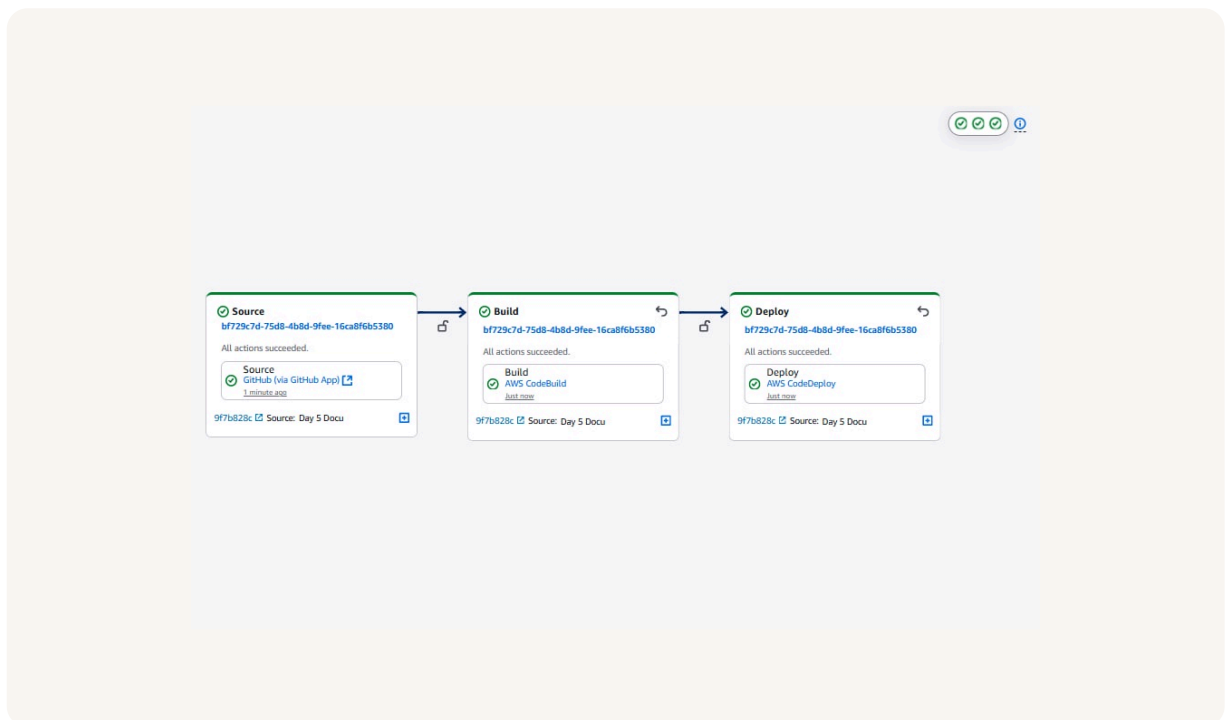




CI/CD Stages

The three stages I've set up in my CI/CD pipeline are Source, Build, Deploy. While setting up each part, I learnt about how the Pipeline is actually setup using Github as the source, CodeBuild as the builder and CodeDeploy as the service that deploys

CodePipeline organizes the three stages into Source, Build and Deploy. In each stage, you can see more details on how the source code is being build and deployed to the webapp.





Source Stage

In the Source stage, the default branch tells CodePipeline which branch to get the source of the changes from.

The source stage is also where you enable webhook events, which are like digital notifications, which CodePipeline automatically start a new pipeline execution in response to this event.

The screenshot shows the 'Add source stage' configuration screen in the AWS CodePipeline console. The interface is divided into a left sidebar with a step list and a main configuration area. The sidebar lists steps 1 through 7, with 'Add source stage' highlighted as Step 3. The main area is titled 'Add source stage' and 'Step 3 of 7'. It contains several sections: 'Source' with a 'Source provider' dropdown set to 'GitHub (via GitHub App)'; 'Connection' with a search bar and a 'Connect to GitHub' button; 'Repository name' with a search bar; 'Default branch' with a search bar; 'Output artifact format' with two radio buttons, 'CodePipeline default' (selected) and 'Full clone'; and 'Webhook events' with a checkbox 'Start your pipeline on push and pull request events' which is checked. There is also a checkbox 'Enable automatic retry on stage failure' which is checked.



Build Stage

The Build stage sets up the deployable build artifact. I configured AWS CodeBuild to compile and package our web application. The input artifact for the build stage is the output from the previous stage that are used as inputs for the current stage.

The screenshot shows the 'Add build stage' configuration screen in the AWS CodePipeline console. The interface is divided into a left sidebar with a step list and a main configuration area. The sidebar lists steps from 'Step 1: Choose creation option' to 'Step 7: Review', with 'Step 4: Add build stage' currently selected. The main area is titled 'Add build stage' and 'Step 4 of 7'. It contains several sections: 'Build provider' with a dropdown set to 'AWS CodeBuild'; 'Project name' with a search field containing '7dayDevOpsChallenge-Day7' and a 'Create project' button; 'Environment variables - optional' with an 'Add environment variable' button; 'Build type' with radio buttons for 'Single build' (selected) and 'Batch build'; 'Region' with a dropdown set to 'Asia Pacific (Singapore)'; 'Input artifacts' with a dropdown set to 'SourceArtifact'; and a checkbox for 'Enable automatic retry on stage failure' which is checked. At the bottom right are buttons for 'Cancel', 'Previous', 'Skip build stage', and 'Next'.



Deploy Stage

The Deploy stage is where the CodePipeline will deploy the build artifact to a webserver. I configured to use my own AWS CodeDeploy as the Deploy provider, and deploy it under the App Name 7DayDevOpsChallenge-Day7

Step 1
Choose creation option

Step 2
Choose pipeline settings

Step 3
Add source stage

Step 4
Add build stage

Step 5
Add test stage

Step 6
Add deploy stage

Step 7
Review

Add deploy stage [info](#)

Step 6 of 7

Deploy - optional

Deploy provider
Choose how you want to deploy your application or content. Choose the provider, and then provide the configuration details for that provider.

AWS CodeDeploy

Region
Asia Pacific (Singapore)

Input artifacts
Choose an input artifact for this action. [Learn more](#)

BuildArtifact
Defined by: Build

No more than 100 characters

Application name
Choose an application that you have already created in the AWS CodeDeploy console. Or create an application in the AWS CodeDeploy console and then return to this task.

7DayDevOpsChallenge-Day7

Deployment group
Choose a deployment group that you have already created in the AWS CodeDeploy console. Or create a deployment group in the AWS CodeDeploy console and then return to this task.

7DayDevOpsChallenge-Day7-deploymentgroup

☒ Configure automatic rollback on stage failure

☐ Enable automatic retry on stage failure

Cancel Previous Skip deploy stage Next



Success!

Since my CI/CD pipeline gets triggered by code changes in source code, I tested my pipeline by adding new lines to index.jsp in my webapp to see if it triggers the pipeline and deploy it to the web.

The moment I pushed the code change, the pipeline started updating from the source till the deploy stage. The commit message under each stage reflects the current stage of the deployment process in the CI/CD pipeline.

Once my pipeline executed successfully, I checked the webapp to see if the changes were applied, and it did get updated.



Hello Amirul!

This is my NextWork webapp working 7!

If you see this line in Github, that means your latest changes are getting pushed to your cloud repo :o

If you see this line, that means your latest changes are automatically deployed into production by CodePipeline!

If you see this line in Github, this means Day 7 of 7 Day DevOps Challenge is Done!!



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